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function [w,wson,uA] =
parabolicbackward(k,h,l,T,alpha)

m = l/h;
n = T/k;

gt0 = @(x,t) 0;           % x=0
gt1 = @(x,t) 0;           % x=l
gx0 = @(x,t) sin(pi*x);  % t=0

x = 0:h:l;
t = 0:k:T;

lambda = (alpha^2)*(k/(h^2));
mu = 1+2*lambda;

% Katsayilar matrisinin parçaları

A = zeros(m-1,m-1);
B = zeros(m-1,m-1);

for i = 1:m-2
    A(i,i+1) = -lambda;
    A(i+1,i) = -lambda;
    A(i,i) = mu;
end
A(m-1,m-1) = mu;

for i = 1:m-1
    B(i,i) = -1;
end

% Katsayilar matrisi

K = zeros((m-1)*n, (m-1)*n);

for i = 1:n
    K(1+(m-1)*(i-1):(m-1)*i, 1+(m-1)*(i-1):(m-1)*i) = A;
end

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for i=1:n-1

    K(1+(m-1)*i:(m-1)*(i+1), 1+(m-1)*(i-1):(m-1)*i) = B;

end

F = zeros(n*(m-1),1);

F(1,1) = gx0(x(2),0) + lambda*gt0(0,t(2));
F(m-1,1) = gx0(x(m),0) + lambda*gtl(0,t(2));

for i = 2:m-1

    F(i,1) = gx0(x(i+1),0);

end

for j = 3:n+1

    F((m-1)*(j-2)+1,1) = lambda*gt0(0,t(j));
    F((m-1)*(j-1),1) = lambda*gtl(0,t(j));

end

% Bilinmeyen vektörü

w = K\F;

wson = w((m-1)*(n-1)+1:(m-1)*n,1);

% Analitik

[T,X] = meshgrid(t,x);

uA = exp(-(pi^2)*T).*sin(pi*X);

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